F1 Podium Prediction — Analytical Report

Auto-generated by the notebook from the executed run. Figures are in `./figures/`.

1. Data & cleaning

- 14 Ergast tables loaded as raw strings; `\N` sentinels → NaN, dates/clock times/lap-times typed (lap times → seconds). Raw CSVs verified unchanged by SHA-256.
- PK/FK audit: see tables in notebook §3 (`figures/04_fk_orphans.png`).
- Grain: 91 duplicate result rows removed (shared drives); labels changing under alt rules: {'best_only': 0, 'worst_only': 21, 'first_resultId': 16}. Rule: prefer grid>0 row, then best positionOrder.
- Modeling window starts in **2003** (first season after which qualifying coverage stays ≥ 90%; see the EDA figures below).

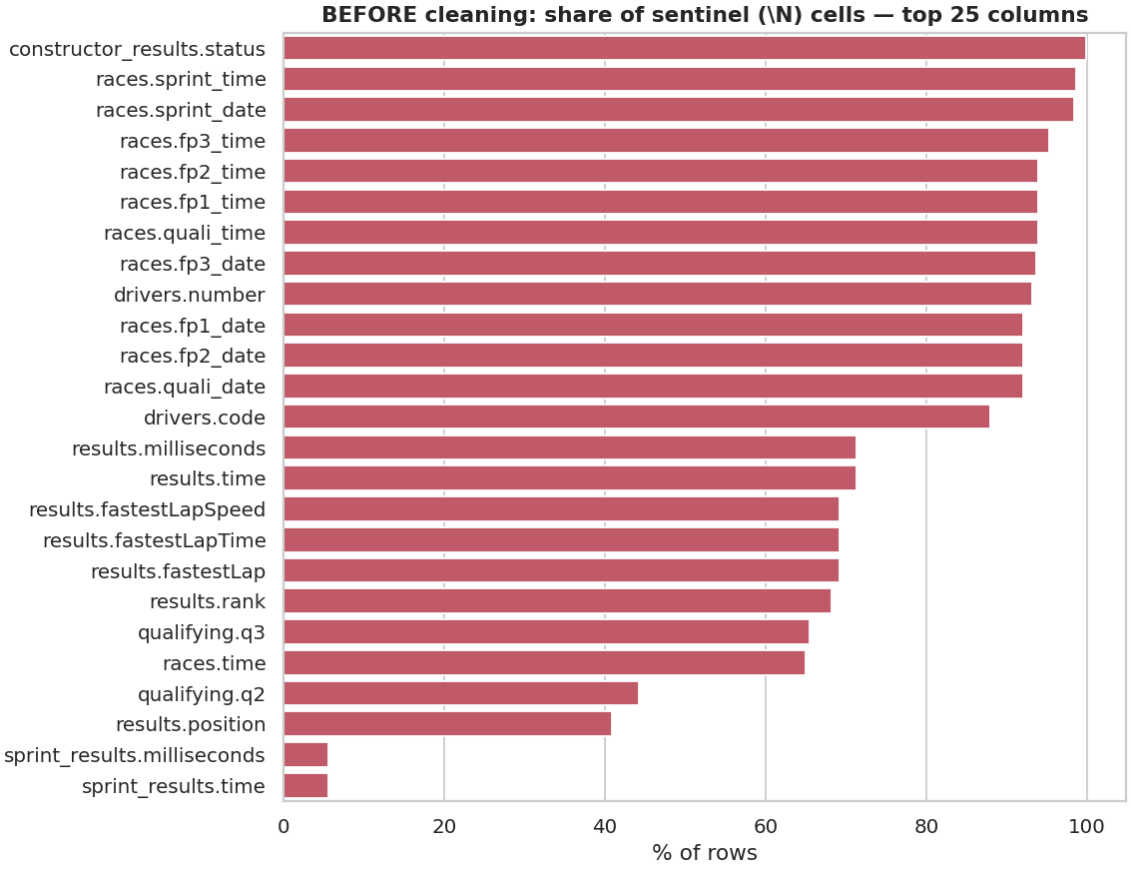


Figure 1 — sentinels before *(01_sentinels_before.png)*

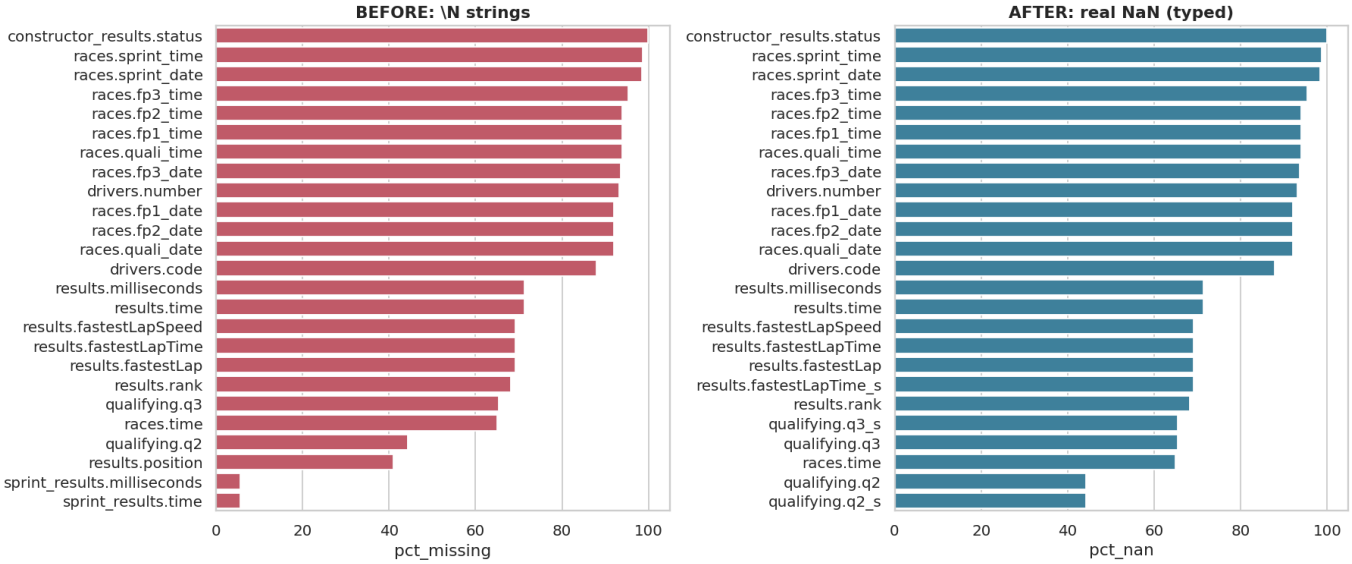


Figure 2 — missing before after *(02_missing_before_after.png)*

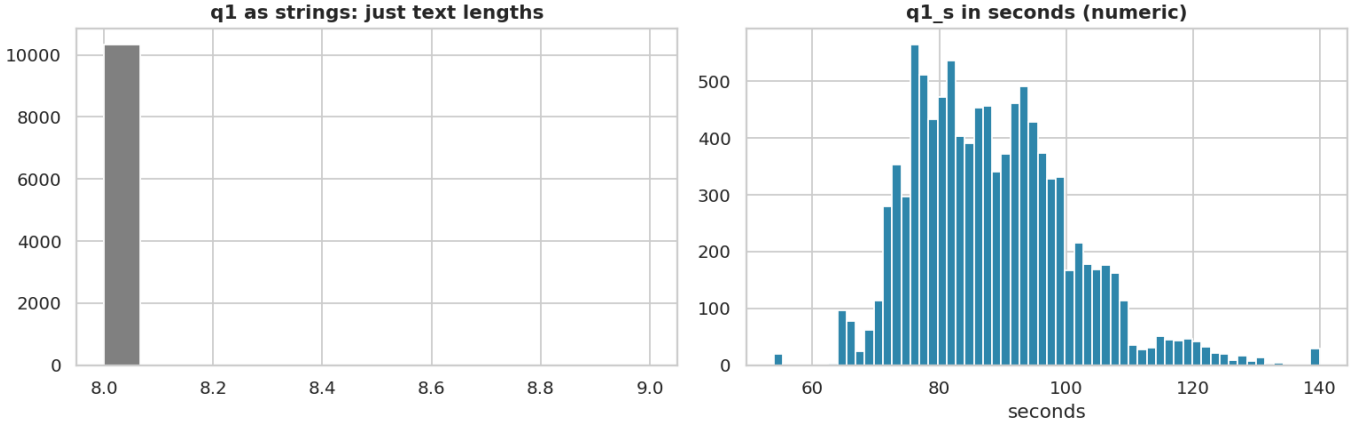


Figure 3 — laptime semantic typing *(03_laptime_semantic_typing.png)*

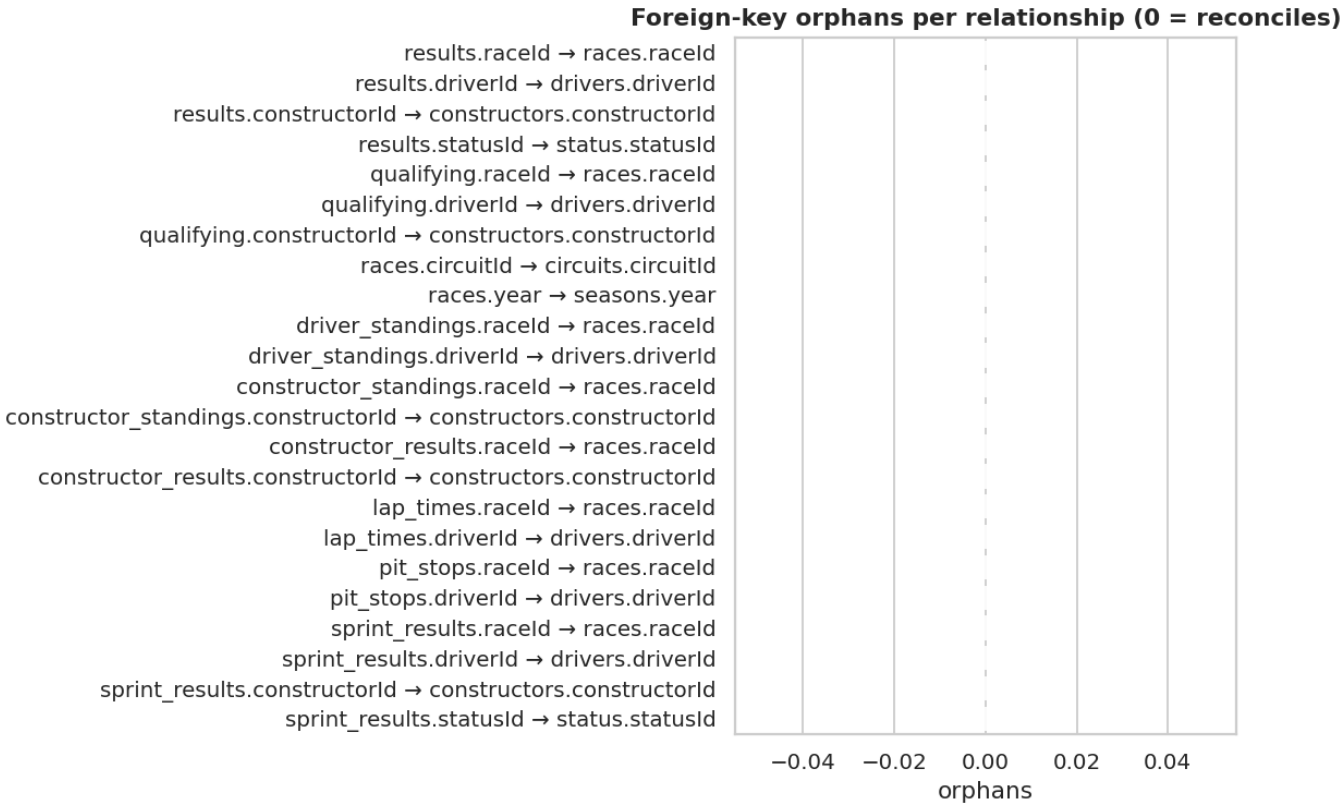


Figure 4 — fk orphans *(04_fk_orphans.png)*

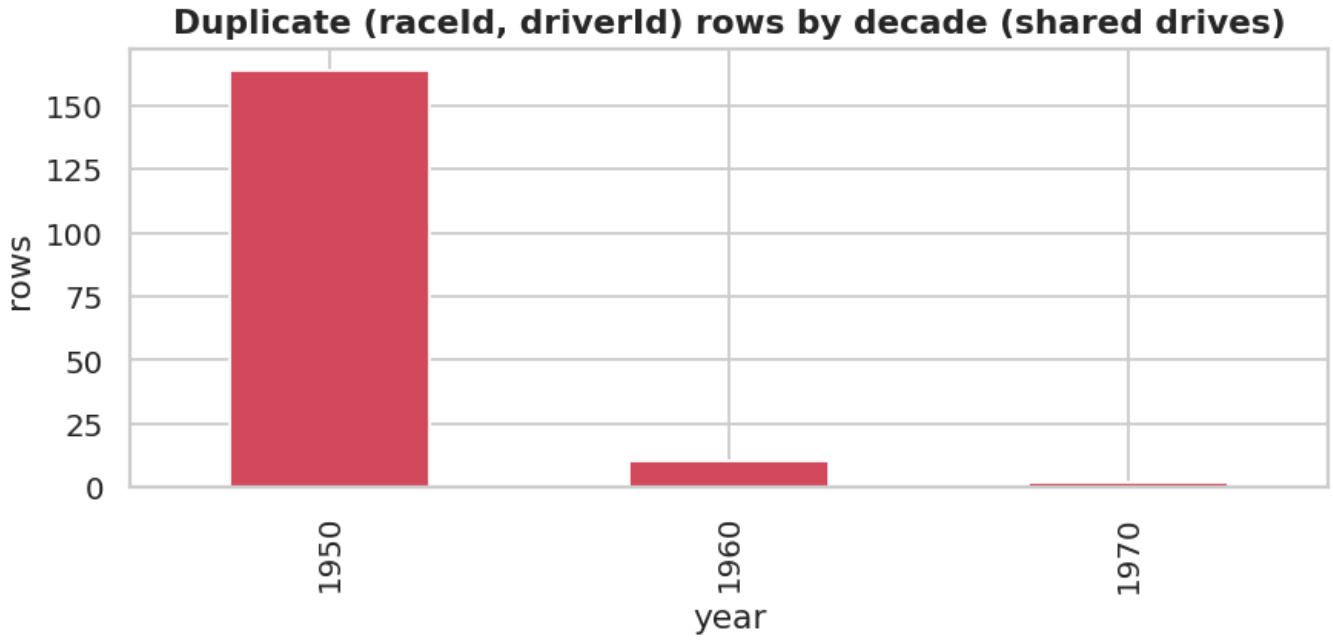


Figure 5 — duplicates by decade *(05_duplicates_by_decade.png)*

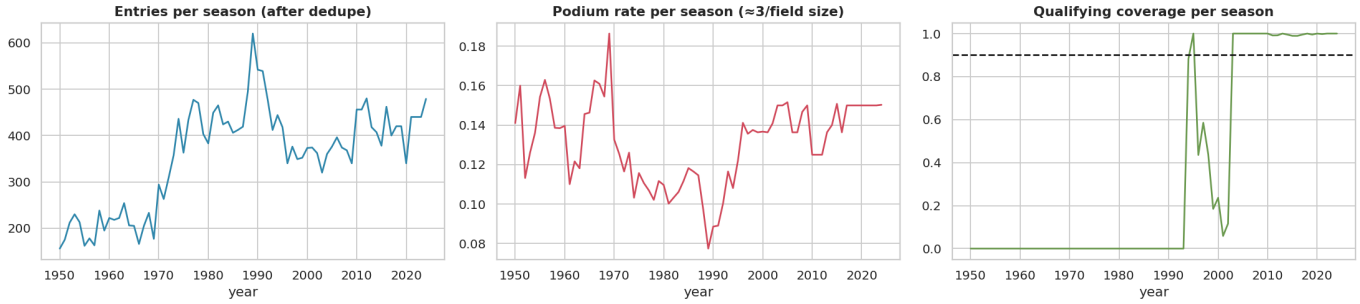


Figure 6 — eda after cleaning *(06_eda_after_cleaning.png)*

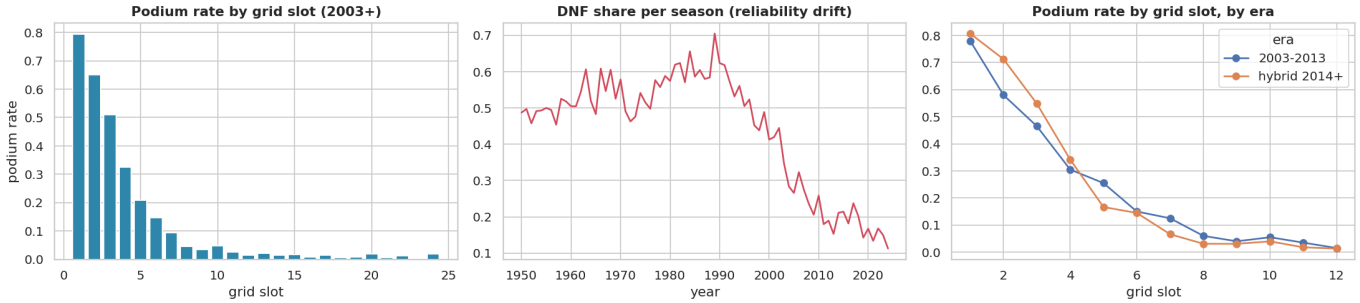


Figure 7 — eda target *(06b_eda_target.png)*

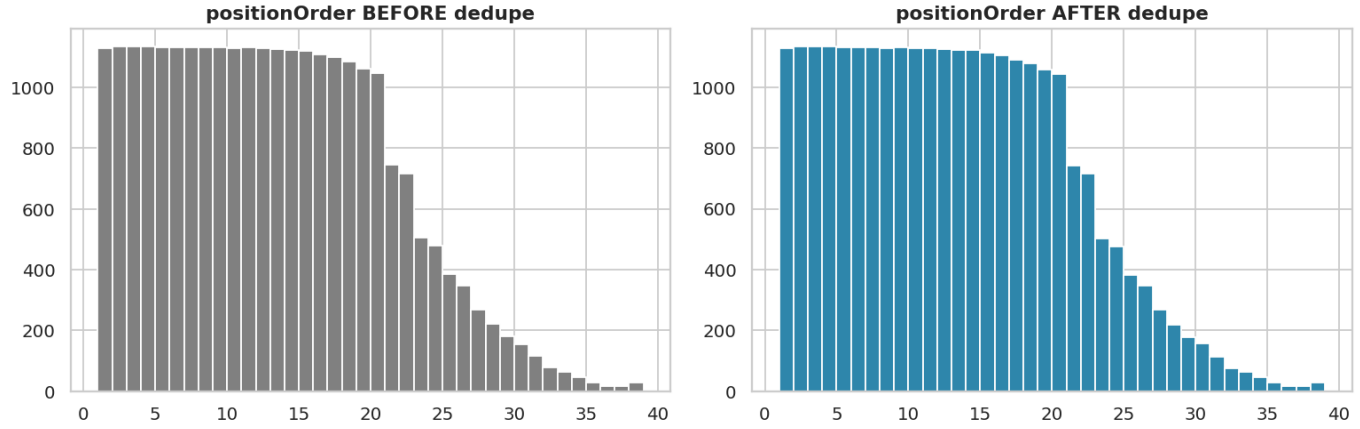


Figure 8 — positionorder before after *(07_positionorder_before_after.png)*

2. Data-engineering answers

Q1 (front row → podium by circuit). Top front-row→podium circuits (≥30 front-row starts) — by qualifying position: ['Suzuka Circuit', 'Circuit de Spa-Francorchamps', 'Circuit de Monaco']; by grid slot: ['Circuit de Monaco', 'Suzuka Circuit', 'Yas Marina Circuit']. Pooled front-row conversion rises from 67% to 74% (qualifying definition) and 68% to 76% (grid definition) in the hybrid era. Rank agreement pre-hybrid vs hybrid: qualifying definition moderate (Spearman ρ=0.57, p=0.033); grid definition weak (Spearman ρ=0.30, p=0.292; not significant). The ranking is only partly stable: it is moderately preserved when the front row is defined by qualifying position but weakly preserved when defined by the penalty-adjusted grid slot, so penalties matter for which circuits look 'good at converting'. Per-circuit samples are small (see Wilson CIs).

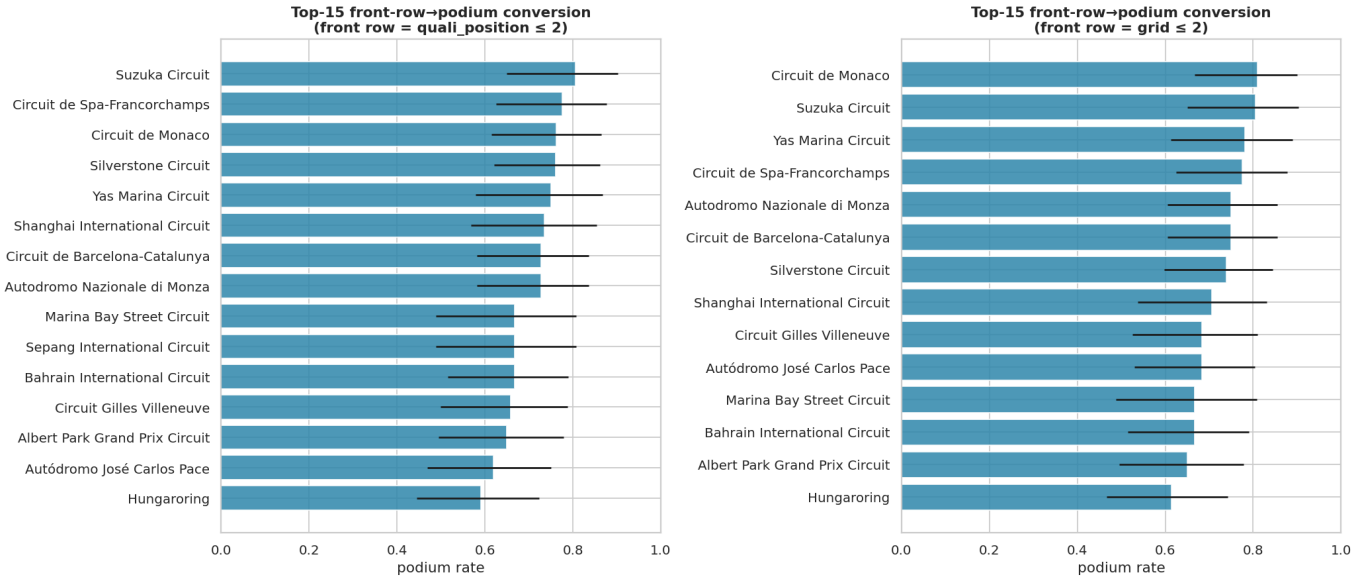


Figure 9 — q1 top circuits *(08_q1_top_circuits.png)*

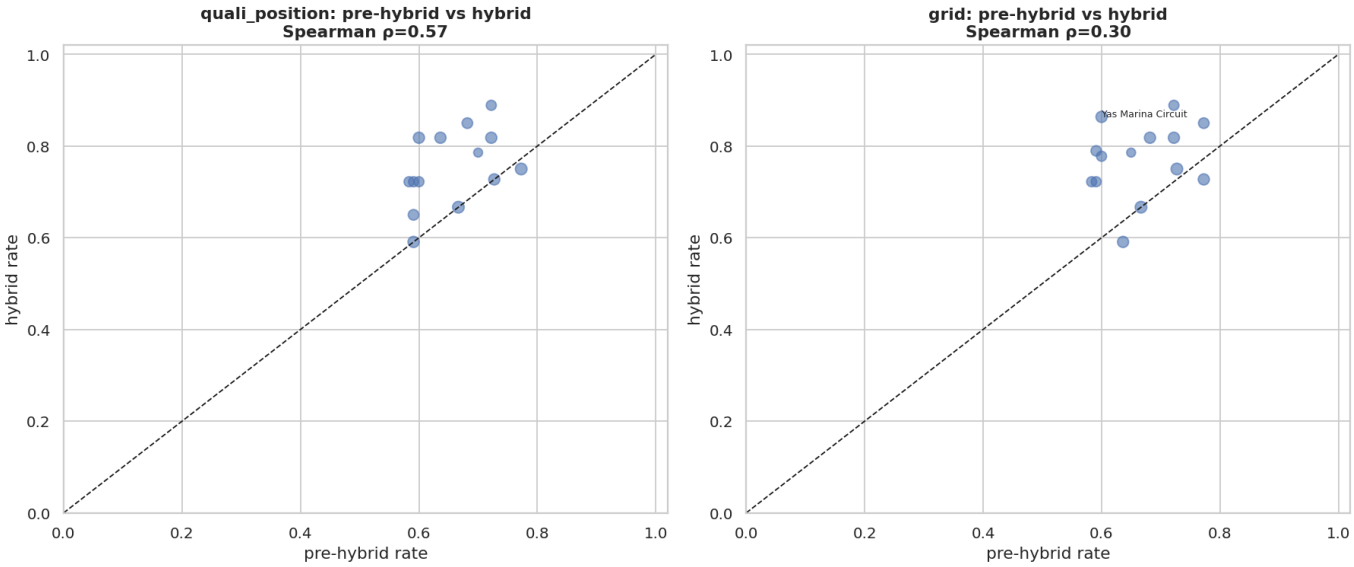


Figure 10 — q1 era change *(09_q1_era_change.png)*

Q2 (home advantage). Unadjusted, home drivers podium with OR=0.81 (home less often). The unadjusted comparison is confounded by starting position: home drivers start further back on the grid on average (mean grid slot 13.4 vs 11.7 for away drivers). After controlling for grid slot the OR is 0.97 (95% CI 0.84–1.12; Mantel–Haenszel 0.97, adding decade fixed effects 0.94); the interval includes 1: no reliable home effect once starting position is held fixed. Caveats: this is an association, not a causal effect, and constructor strength is not controlled.

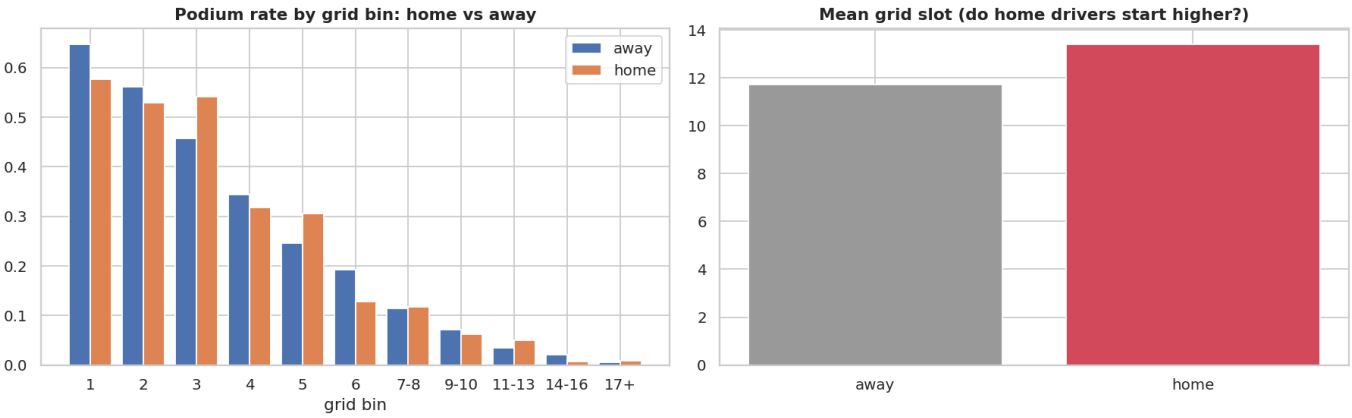


Figure 11 — q2 home vs away *(10_q2_home_vs_away.png)*

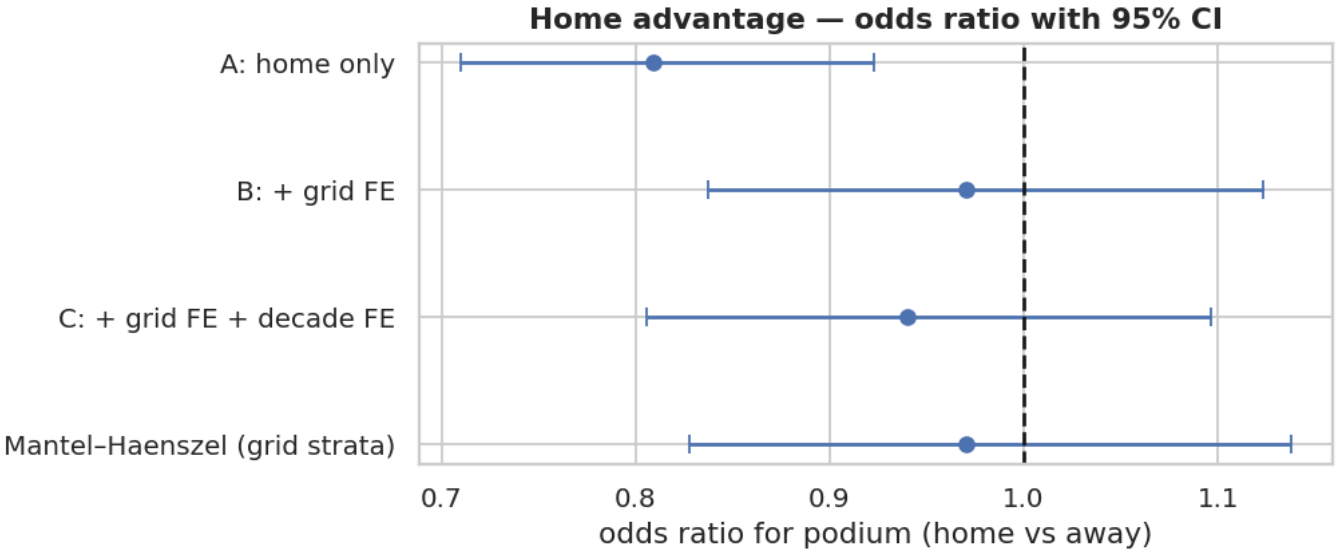


Figure 12 — q2 forest *(11_q2_forest.png)*

Q3 (mechanical retirements). Highest mechanical-retirement rate 2014–2021: **AlphaTauri/RB (Toro Rosso)** (14.4%); 2022–2024: **Alpine (Renault/Lotus)** (11.0%). Field-wide rate moved from 9.3% to 5.5% (χ^2 p=<0.001). Largest increase: Sauber/Alfa Romeo (+0.7%); largest decrease: McLaren (-10.2%). 2022–24 ranking is robust to a broader definition (Spearman 0.96). Small samples per team: read with the CIs.

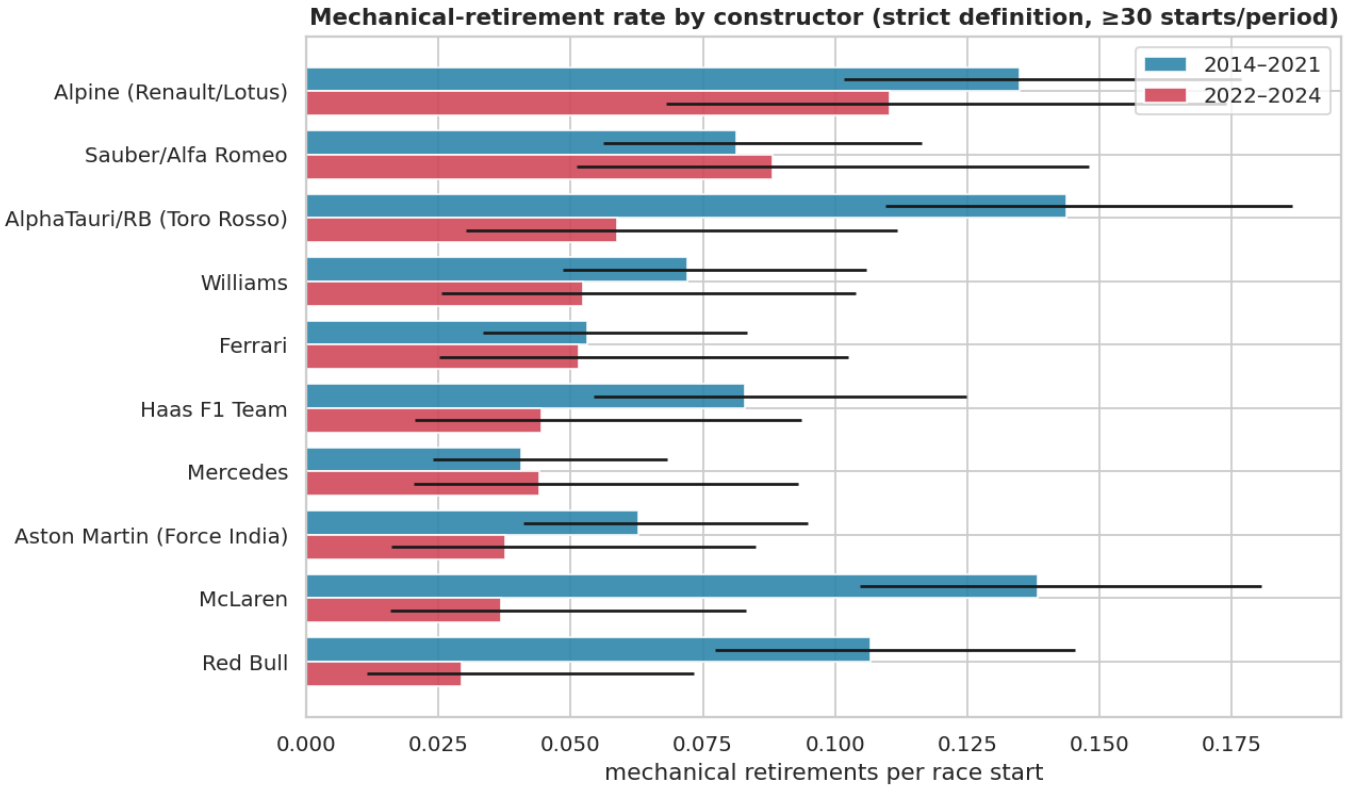


Figure 13 — q3 mech rate *(12_q3_mech_rate.png)*

3. Features and why they are valid at the pre-race cutoff

group	features
grid_quali	grid_filled, is_pit_start, front_row, quali_position, quali_missing, q1_gap_pct, best_gap_pct, grid_minus_quali, grid_penalty
constructor_form	team_form_pos_5, team_form_podium_5, team_form_points_5, team_no_history
standings	drv_pts_ratio, drv_pos_pre, drv_wins_pre, con_pos_pre, con_pts_ratio, standings_missing, con_standings_missing
circuit_history	circ_starts, circ_avg_pos, circ_podium_rate
home	home_race

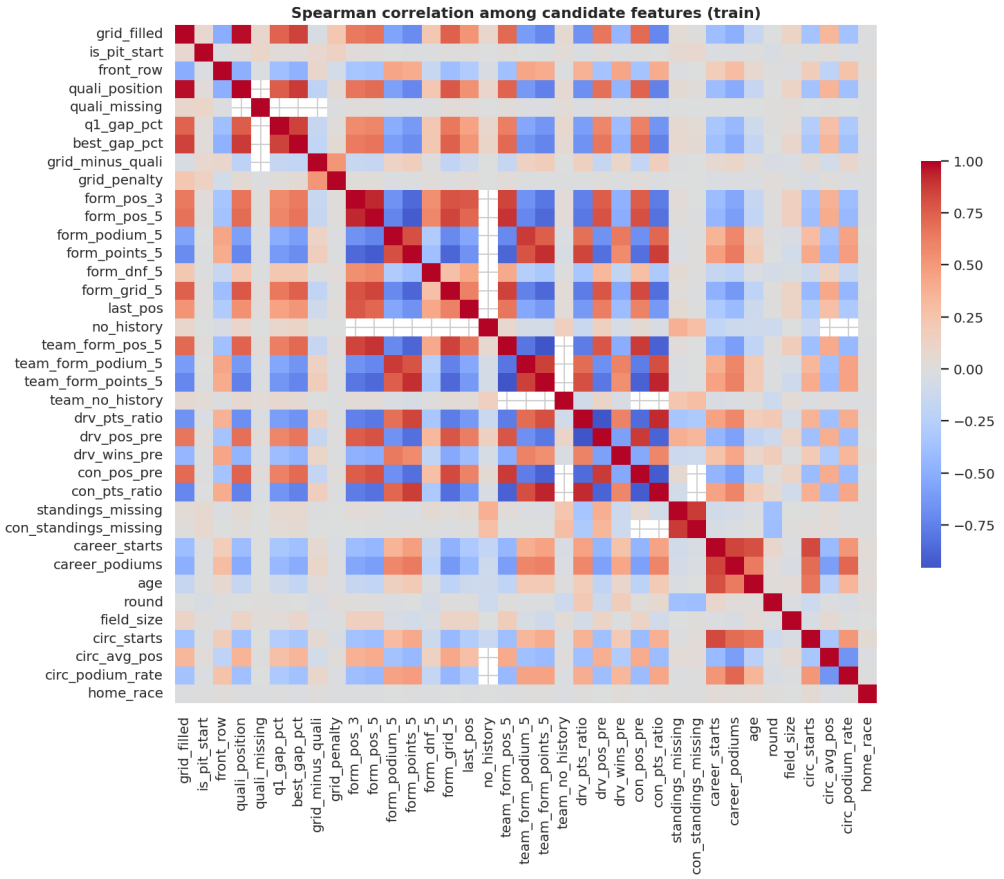


Figure 14 — feature correlation *(15_feature_correlation.png)*

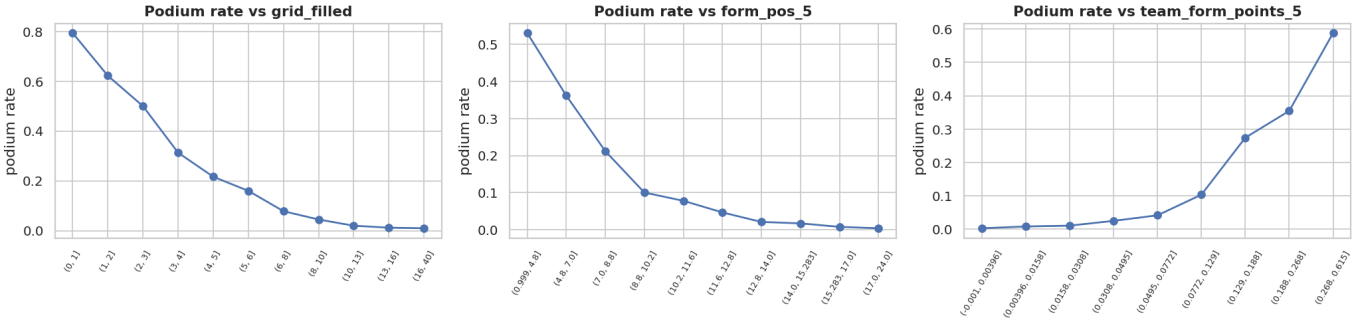


Figure 15 — feature effect on target *(16_feature_effect_on_target.png)*

Form features use `shift(1)` before rolling (past races only); standings come from the previous round of the same season; grid/qualifying are final before the start. Verified by the executable leakage tests (§6.1: manual recompute, current-race perturbation, standings-source ordering).

3b. Feature selection

Groups dropped (validation PR-AUC does not suffer without them, mean Δ ≥ 0.005): ['experience_context', 'driver_form']. Redundancy pruning not applied (mean Δ -0.0007, tolerance 0.005). Final feature set: 24 of 37 candidates.

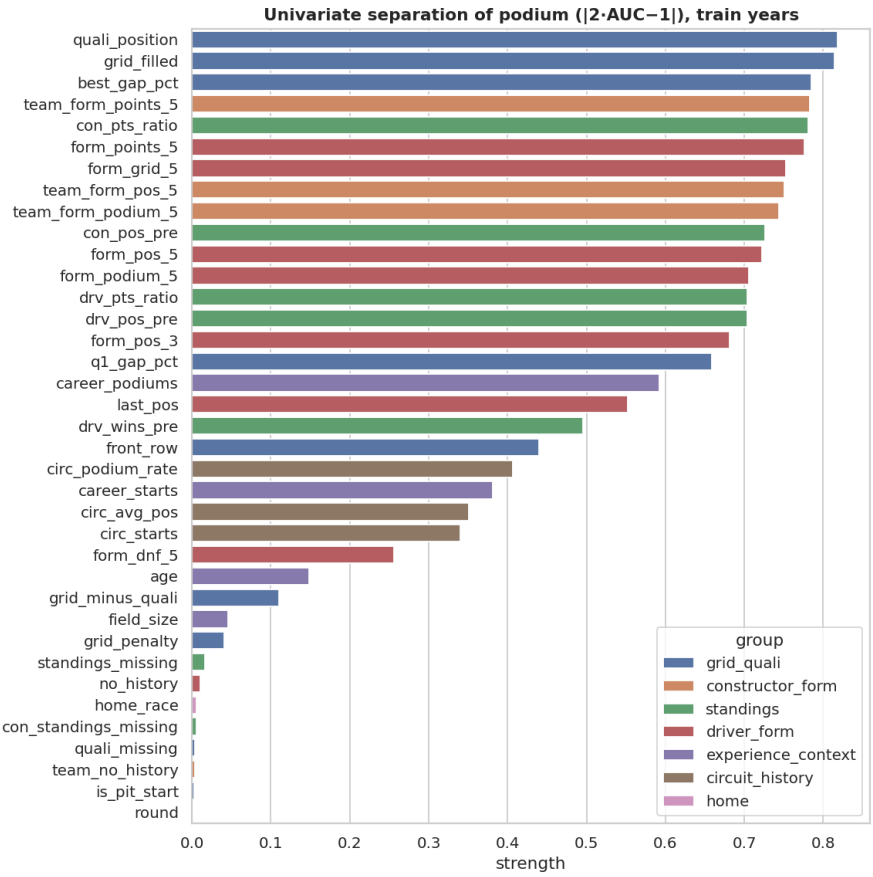


Figure 16 — univariate strength *(14_univariate_strength.png)*

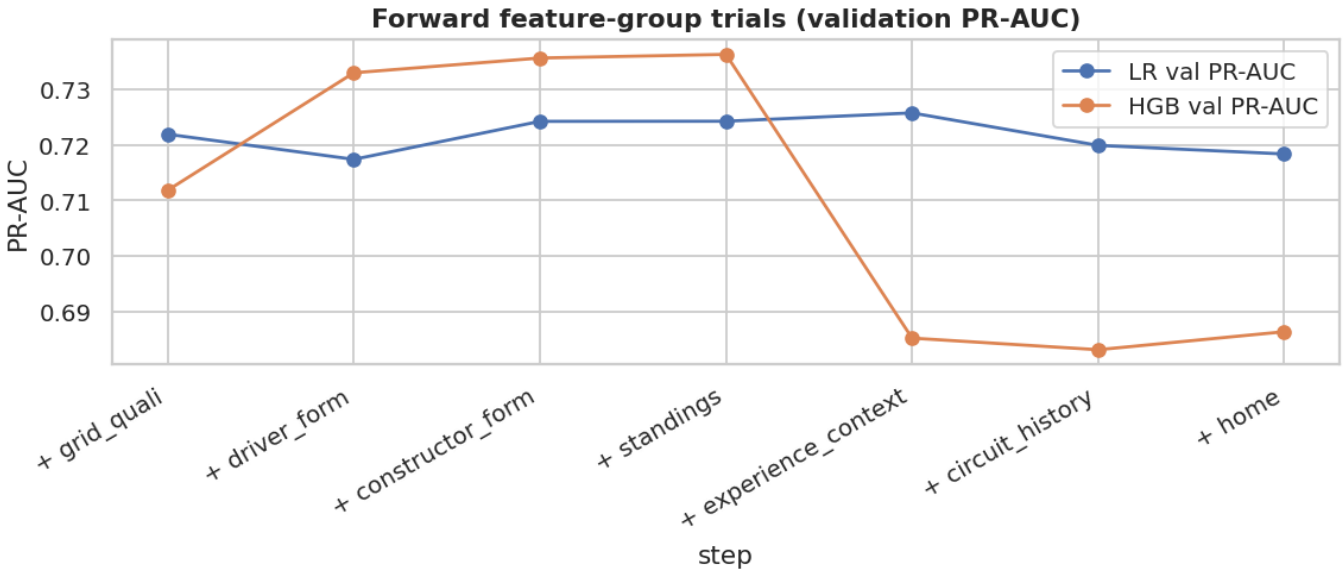


Figure 17 — forward selection *(17_forward_selection.png)*

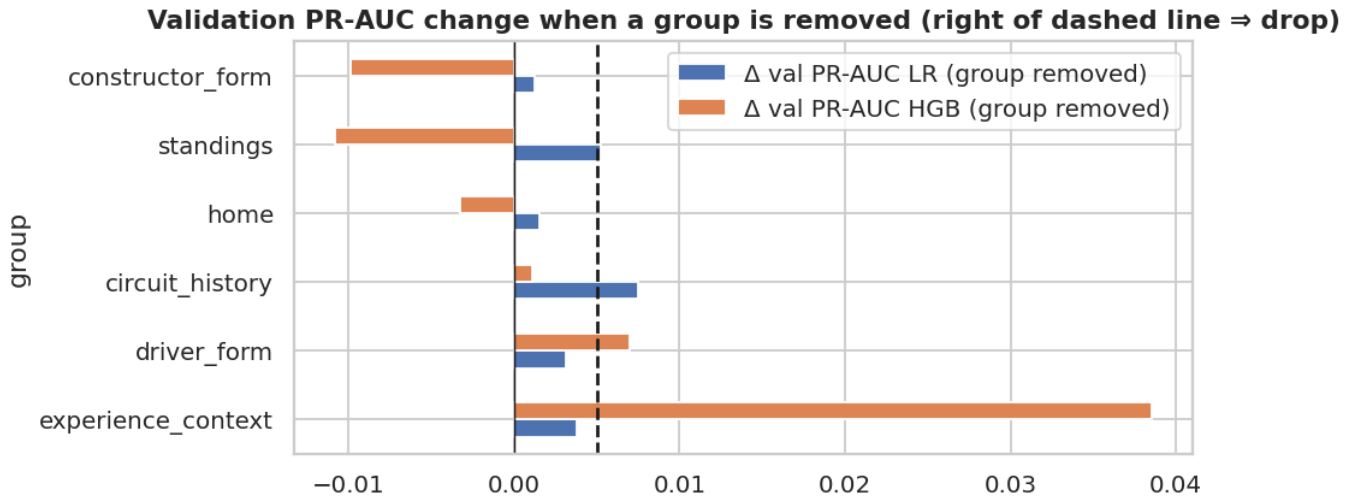


Figure 18 — group pruning *(17b_group_pruning.png)*

4. Split

Chronological: train 2003–2018, validation [2019, 2020, 2021], test [2022, 2023, 2024]. Hyper-parameters and F1 threshold from validation only.

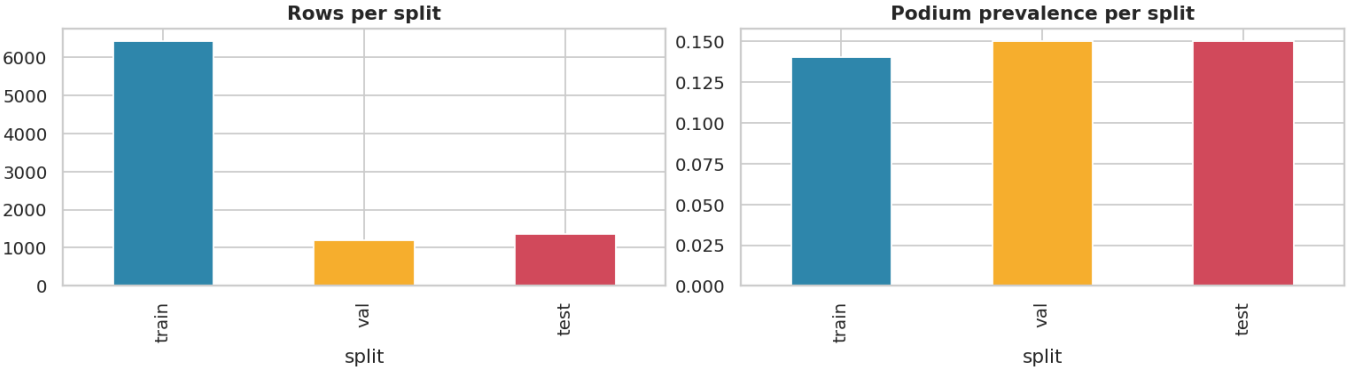


Figure 19 — split *(13_split.png)*

5. Models (primary metric PR-AUC; secondary ROC-AUC and F1)

model	F1 (test)	F1 (val)	PR-AUC (test)	PR-AUC (val)	Precision (test)	Precision (val)	ROC-AUC (test)	ROC-AUC (val)	Recall (test)	Recall (val)
Gradient boosting	0.647	0.730	0.694	0.745	0.652	0.717	0.938	0.940	0.642	0.744
LR baseline (C=1)	0.658	0.722	0.727	0.724	0.534	0.631	0.938	0.934	0.858	0.844
LR grid-only	0.670	0.718	0.682	0.722	0.595	0.652	0.920	0.925	0.765	0.800
LR tuned	0.659	0.722	0.727	0.726	0.535	0.634	0.938	0.934	0.858	0.839
Random forest	0.686	0.731	0.714	0.741	0.636	0.684	0.942	0.939	0.745	0.783
Reference: grid-position rule	0.649	0.717	0.647	0.676	0.650	0.717	0.914	0.917	0.647	0.717
Shallow NN (1 hidden layer)	0.658	0.731	0.710	0.723	0.615	0.690	0.937	0.933	0.706	0.778
Shallow NN (2 hidden layers)	0.657	0.732	0.704	0.729	0.652	0.720	0.935	0.935	0.662	0.744

Final model **Gradient boosting** (best validation PR-AUC). Test PR-AUC 0.694 vs baseline LR 0.727; grid-rule reference 0.647.

Model comparison. Final model **Gradient boosting** (best validation PR-AUC) reaches test PR-AUC 0.694 versus 0.727 for the logistic-regression baseline (Δ=-0.033) and 0.704 for the shallow NN; the grid-position rule

alone reaches 0.647. The NN's validation PR-AUC varies by ±0.003 across seeds, so a gap of 0.033 between FINAL and the baseline is above seed-level noise: the more complex model gives a real gain over the baseline. Tree ensembles over-fit the training seasons more (train PR-AUC ≫ validation), while LR and the NN show a small train→val gap.

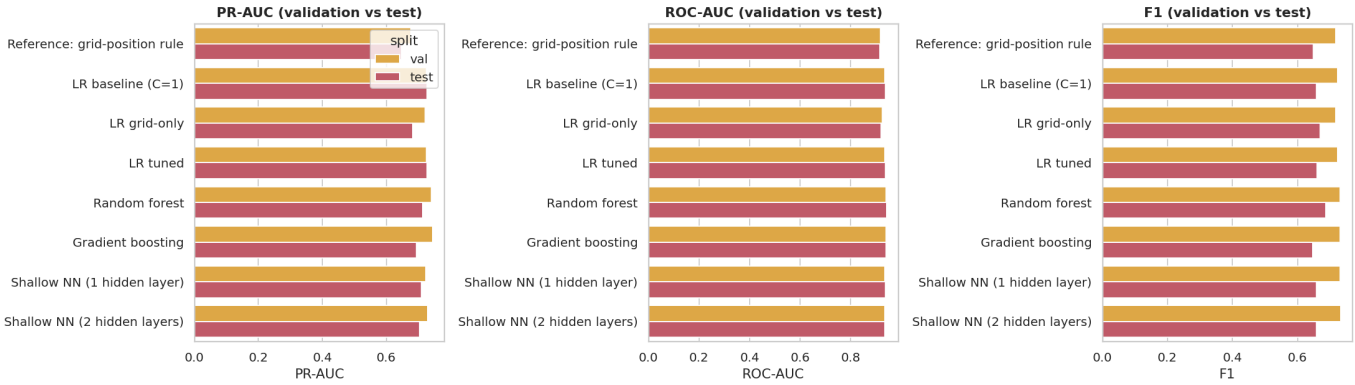


Figure 20 — model comparison *(19_model_comparison.png)*

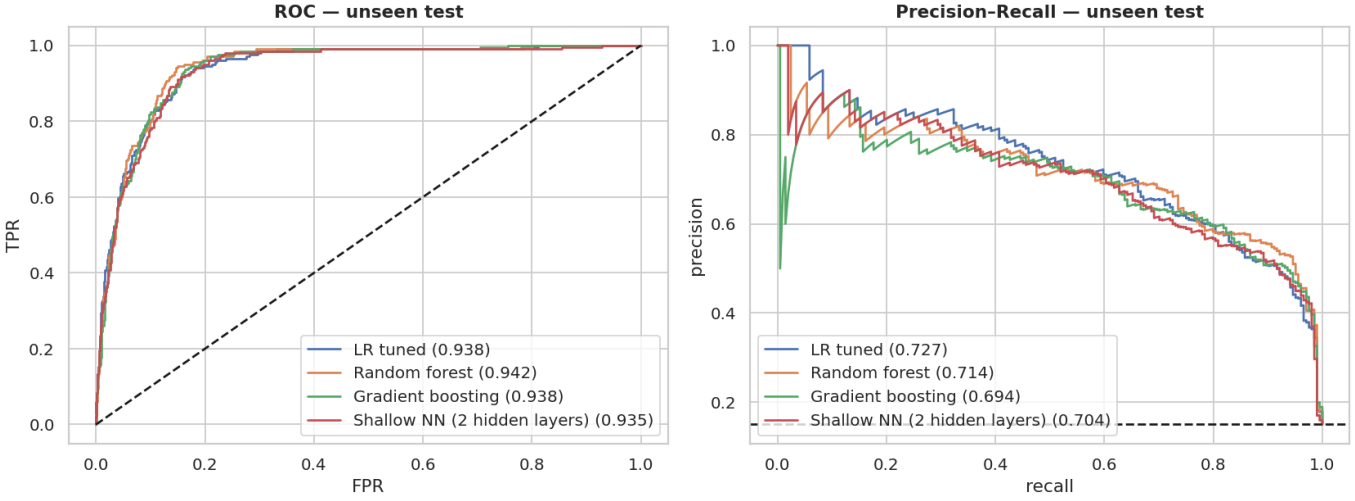


Figure 21 — roc pr test *(20_roc_pr_test.png)*

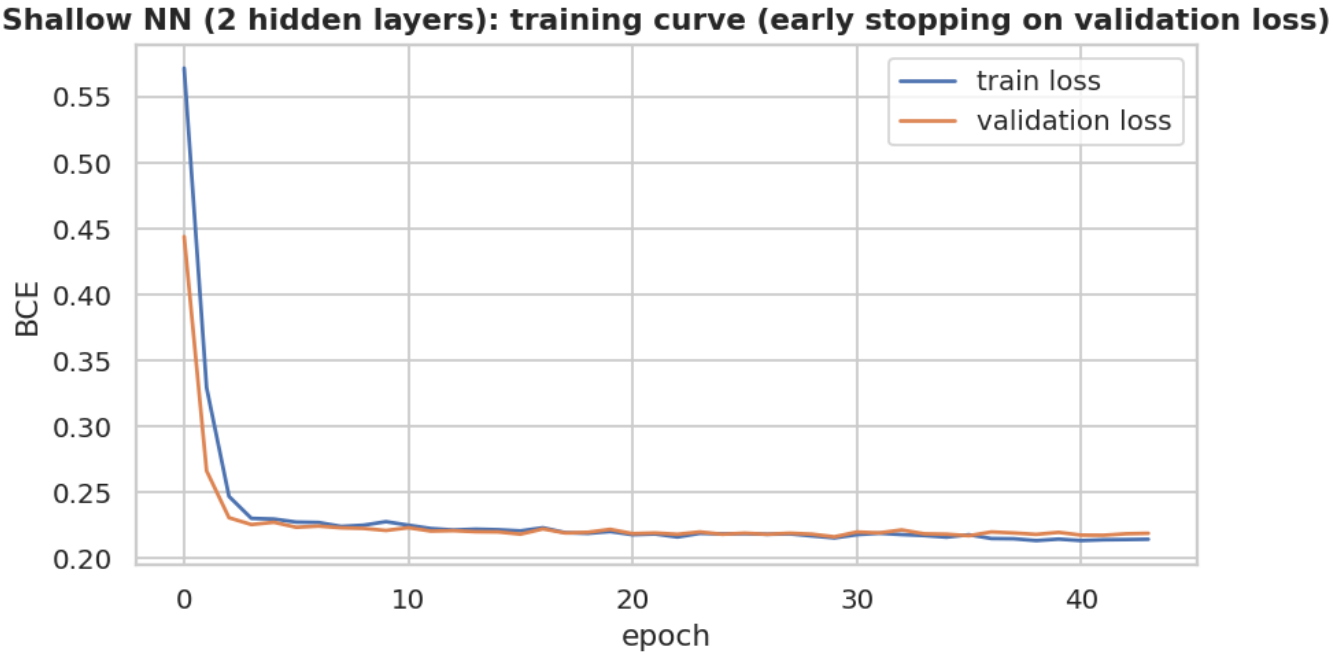


Figure 22 — nn training curve *(18_nn_training_curve.png)*

6. Ablation and experiments

LR: most valuable group − standings (-0.010 val PR-AUC when removed), least valuable − constructor_form (+0.004) | RF: most valuable group − qualifying (-0.018 val PR-AUC when removed), least valuable − constructor_form (+0.001) | HGB: most valuable group − standings (-0.021 val PR-AUC when removed), least valuable − circuit_history (+0.001) | NN: most valuable group − standings (-0.004 val PR-AUC when removed), least valuable − circuit_history (+0.007)

exp	variant	val PR-AUC	test PR-AUC	test ROC-AUC
A	fit on TRAIN only (correct)	0.7259	0.7271	0.9379
A	fit on ALL data (leaky)	0.7236	0.7281	0.9380
B	winsor → impute → scale	0.7259	0.7271	0.9379
B	impute → winsor → scale	0.7258	0.7271	0.9379
B	impute → scale → winsor	0.7256	0.7279	0.9382
C	engineer on full history, then filter (chosen)	0.7259	0.7271	0.9379
C	filter first, then engineer	0.7261	0.7285	0.9380
D	chronological split (chosen)	0.7259	0.7271	0.9379
D	random split (same sizes)	0.7117	0.6887	0.9143
E	train from 2003 (chosen)	0.7259	0.7271	0.9379
E	train from 1950 (all seasons)	0.7333	0.7250	0.9363

EXP-E: training from 1950 changes validation PR-AUC by +0.0074 and test PR-AUC by -0.0021 versus training from 2003. The gain is larger than seed noise and the test gain is negligible. We keep 2003 because qualifying times are largely absent earlier (those rows rely on median imputation + missing-flags), the pre-2003 cars/regulations are far from the 2022-24 test regime, and the data window then matches the task definition ('after qualifying'); we acknowledge that more history did not hurt and may help slightly.

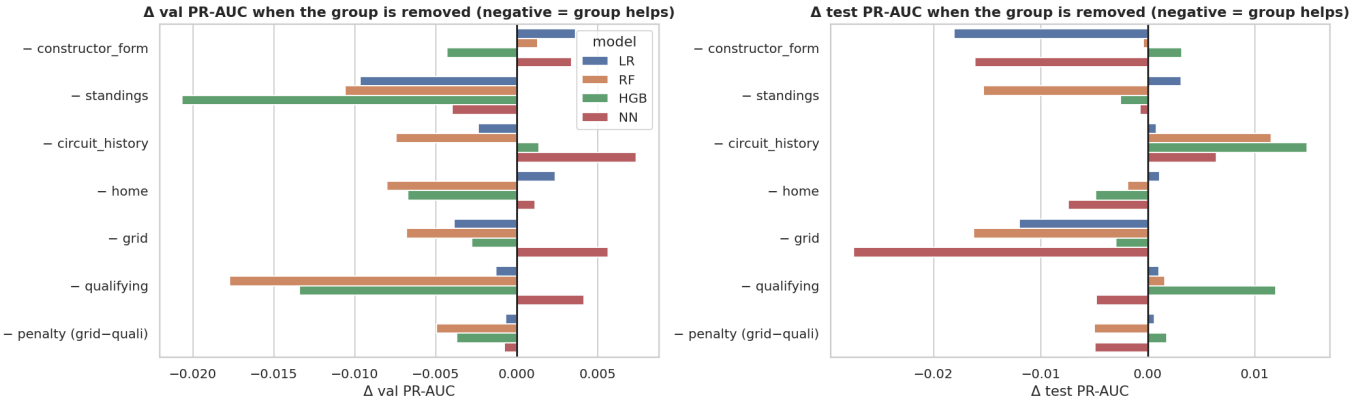


Figure 23 — ablation *(21_ablation.png)*

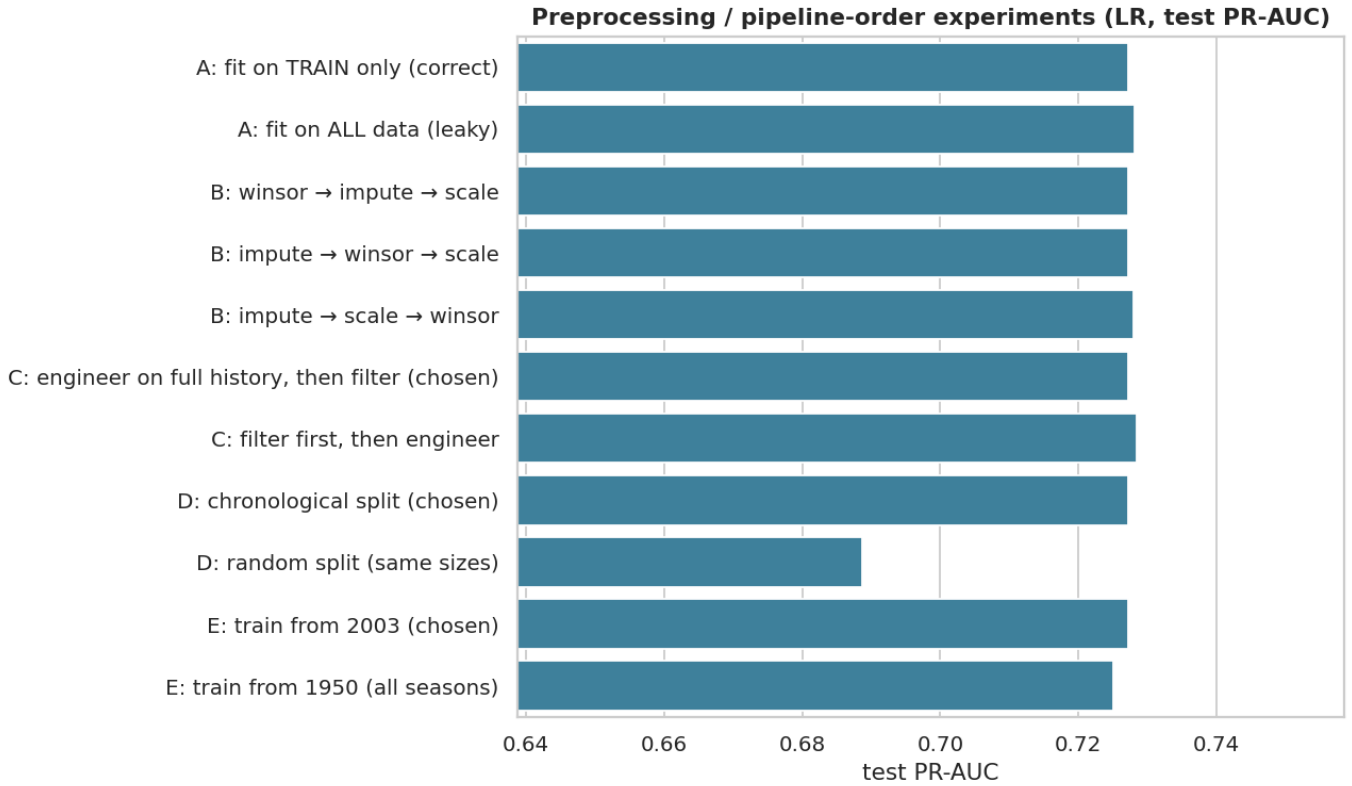


Figure 24 — preprocessing experiments *(22_preprocessing_experiments.png)*

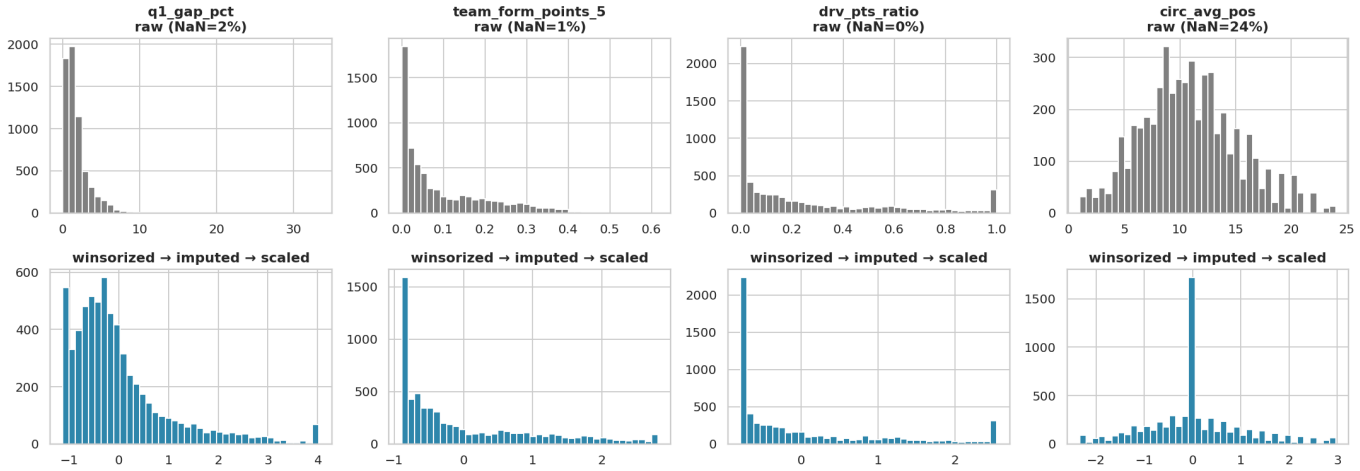


Figure 25 — preprocessing distribution *(23_preprocessing_distribution.png)*

7. Explainability

Permutation importance top-5: ['grid_filled', 'quali_position', 'team_form_points_5', 'drv_wins_pre', 'q1_gap_pct']; SHAP top-5: ['team_form_points_5', 'quali_position', 'best_gap_pct', 'grid_filled', 'team_form_pos_5']. Rank agreement over all features ρ=0.57; top-10 overlap = 8/10 (['grid_filled', 'quali_position', 'team_form_points_5', 'drv_wins_pre', 'con_pts_ratio', 'best_gap_pct', 'team_form_pos_5', 'drv_pos_pre']). Only in permutation top-10: ['q1_gap_pct', 'home_race']; only in SHAP top-10: ['circ_podium_rate', 'drv_pts_ratio']. Likely reason for the disagreement — correlated feature pairs: drv_pts_ratio~team_form_points_5 (ρ=0.82); drv_pts_ratio~con_pts_ratio (ρ=0.95); drv_pts_ratio~team_form_pos_5 (ρ=-0.80); drv_pts_ratio~drv_pos_pre (ρ=-0.97). Permuting one member of a pair leaves its partner intact, so permutation importance can under-state both, while SHAP spreads credit between them; grouped permutation (below) is therefore the fairer group-level view. Grouped permutation ranks the groups ['grid_quali', 'standings', 'constructor_form', 'home', 'circuit_history'] (PR-AUC drop [0.358, 0.04, 0.031, 0.003, -0.012]). Direction of the top SHAP features: team_form_points_5 (higher value → higher podium probability); quali_position (higher value → lower podium probability); best_gap_pct (higher value → lower podium probability); grid_filled (higher value → lower podium probability); team_form_pos_5 (higher value → lower podium probability). **These are model-reliance explanations, not causal effects.**

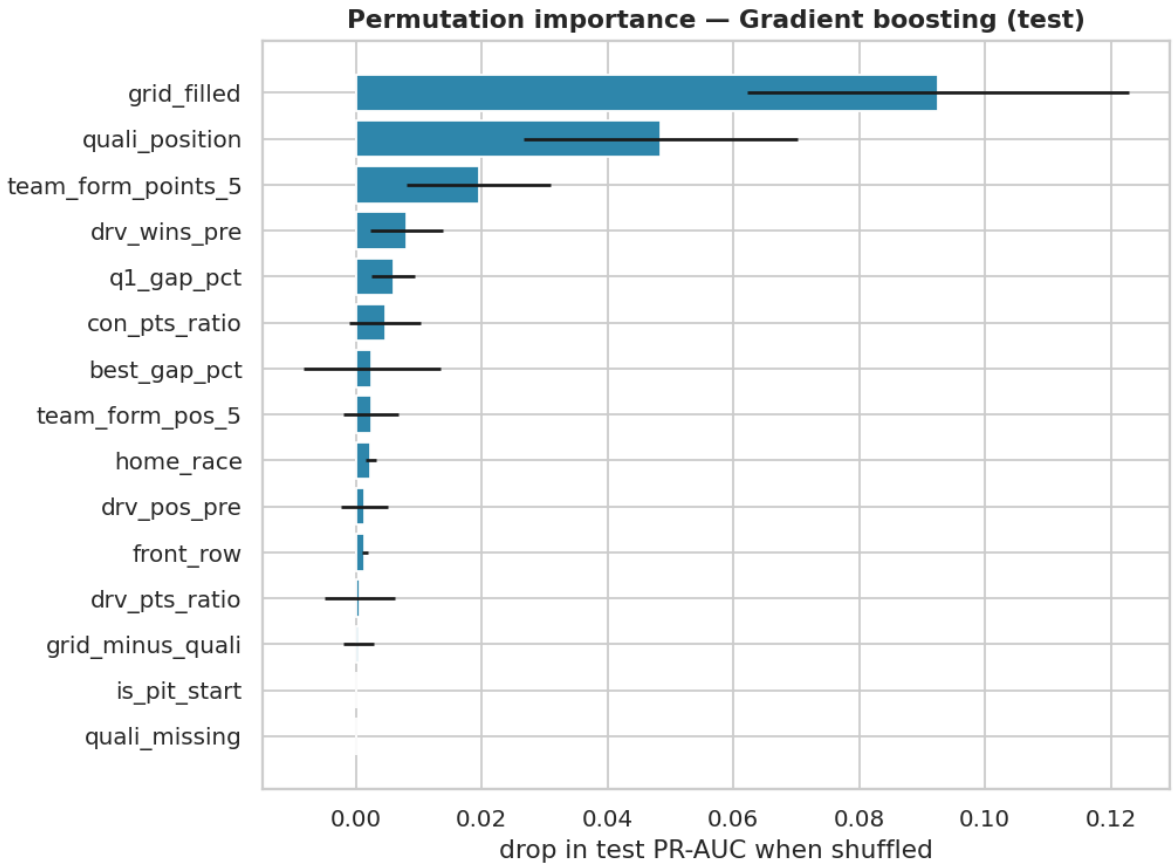


Figure 26 — permutation importance *(24_permutation_importance.png)*

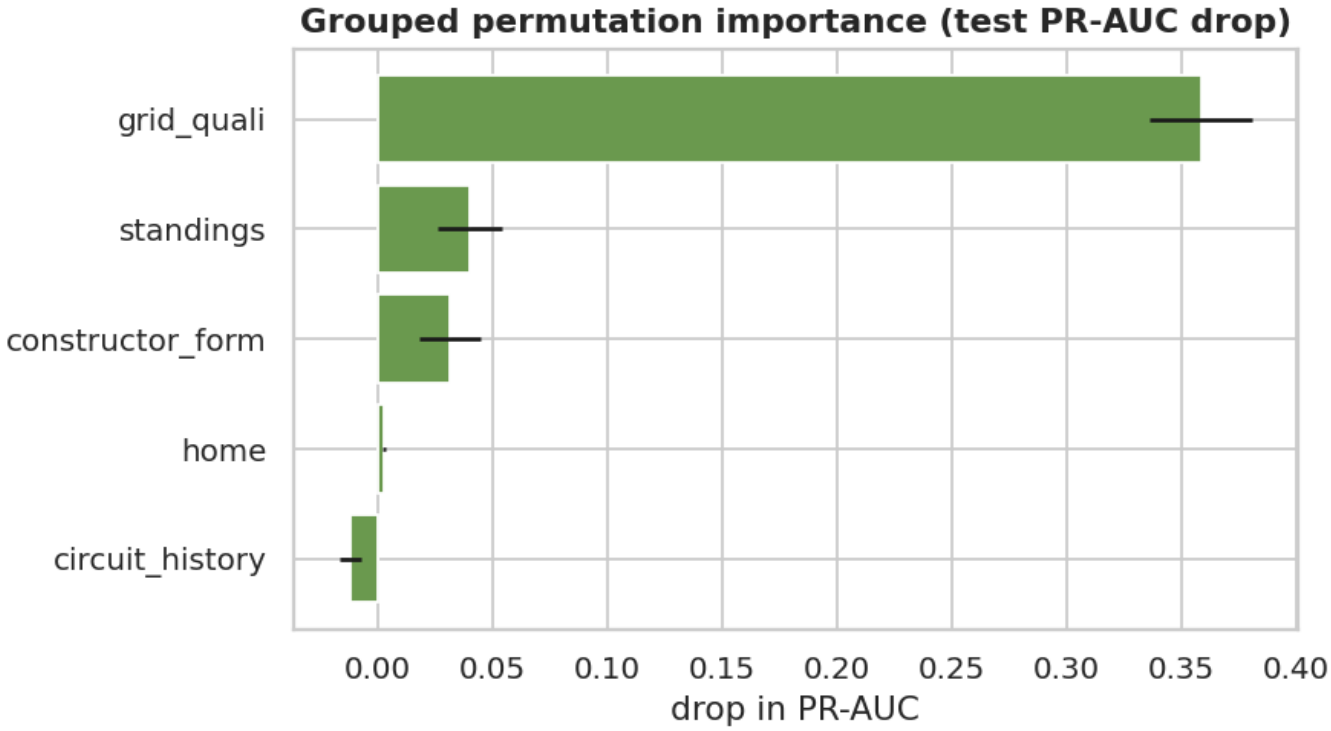


Figure 27 — grouped permutation *(25_grouped_permutation.png)*

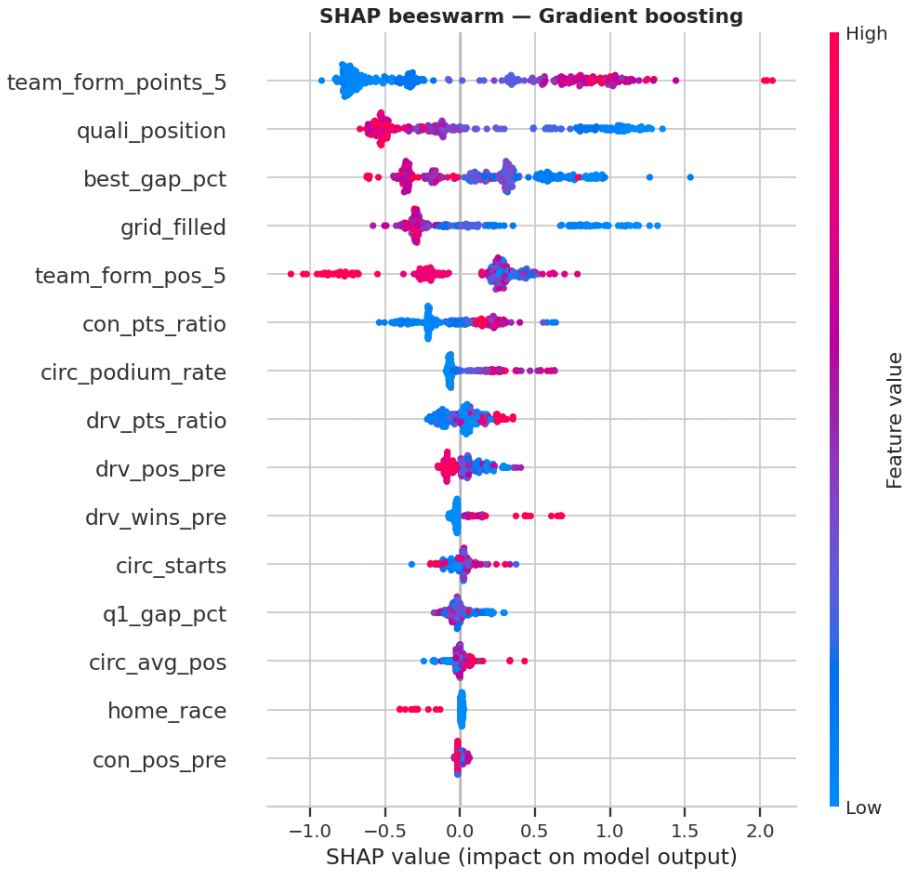


Figure 28 — shap beeswarm *(26_shap_beeswarm.png)*

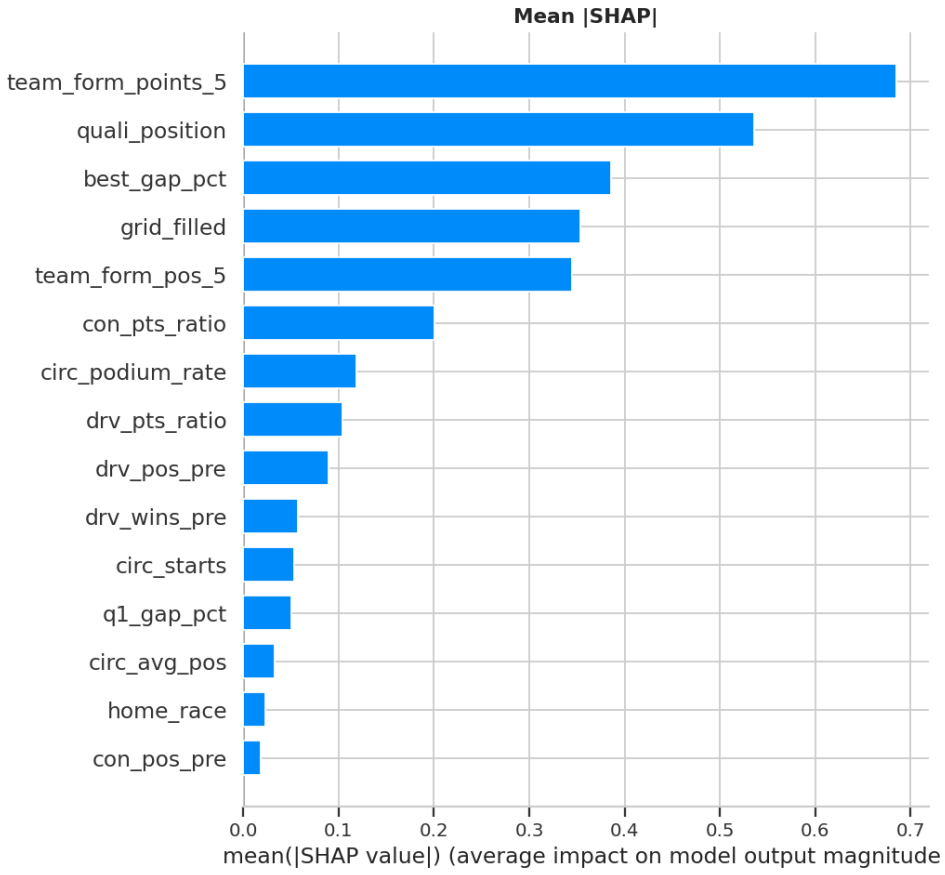


Figure 29 — shap bar *(27_shap_bar.png)*

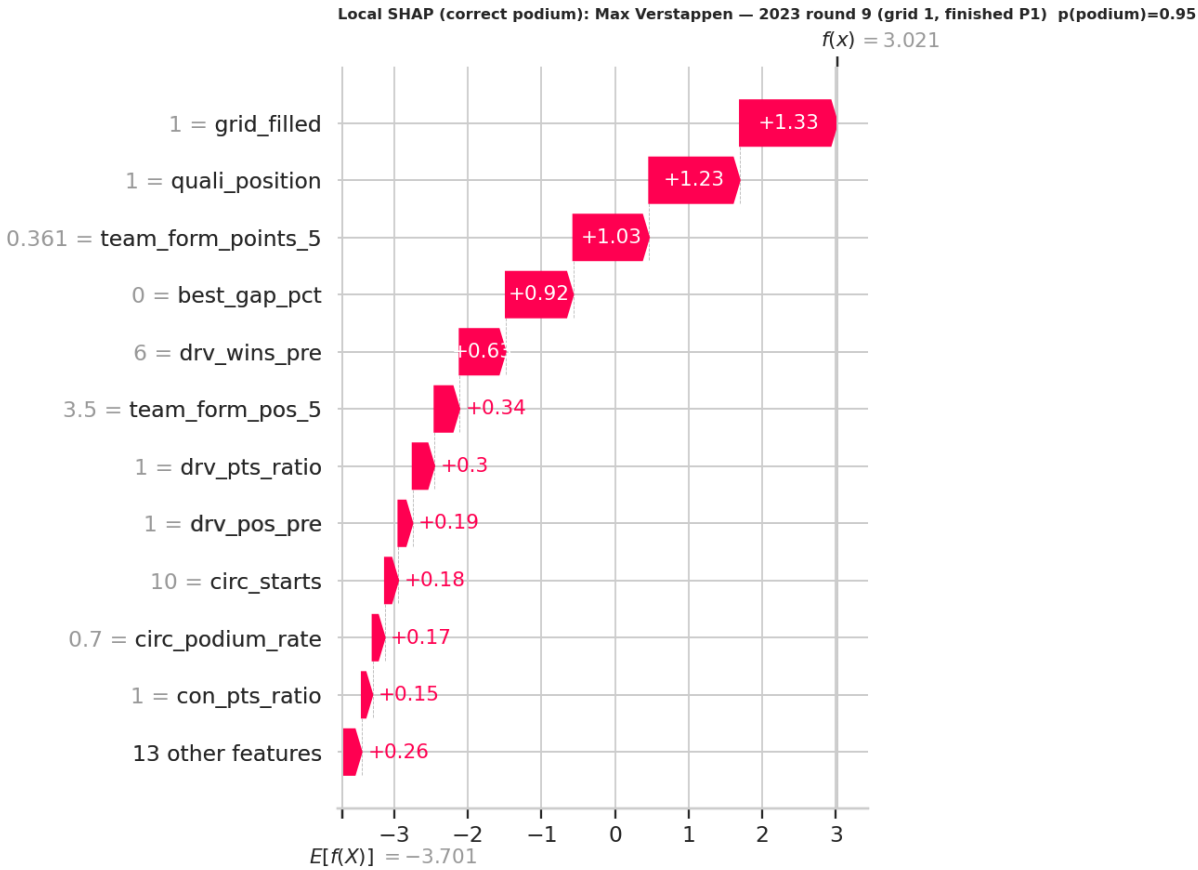


Figure 30 — shap waterfall correct *(28_shap_waterfall_correct.png)*

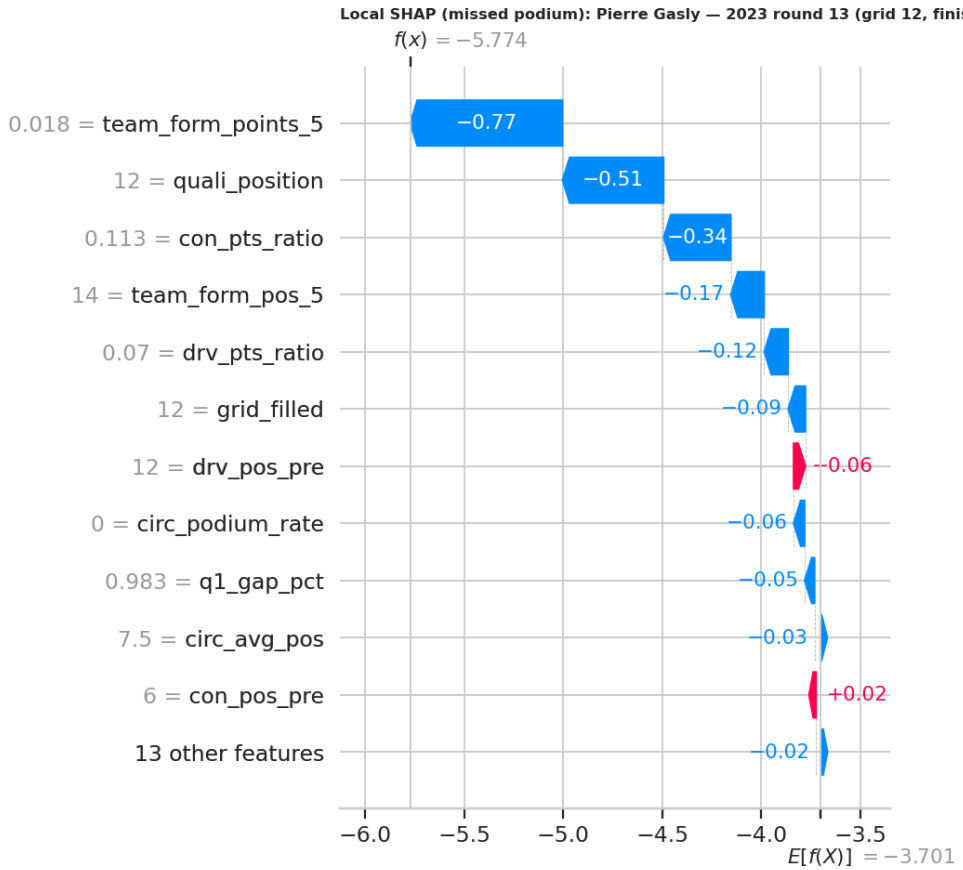


Figure 31 — shap waterfall missed *(28_shap_waterfall_missed.png)*

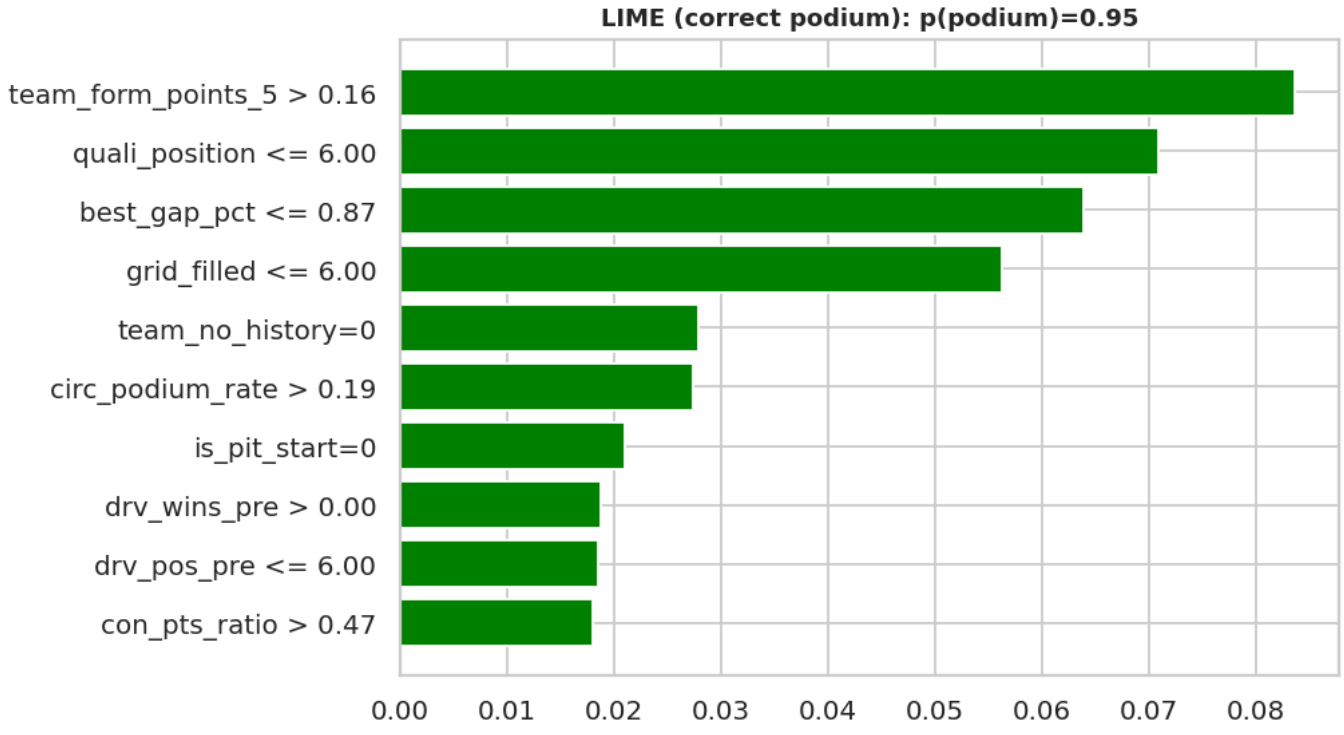


Figure 32 — lime correct *(29_lime_correct.png)*

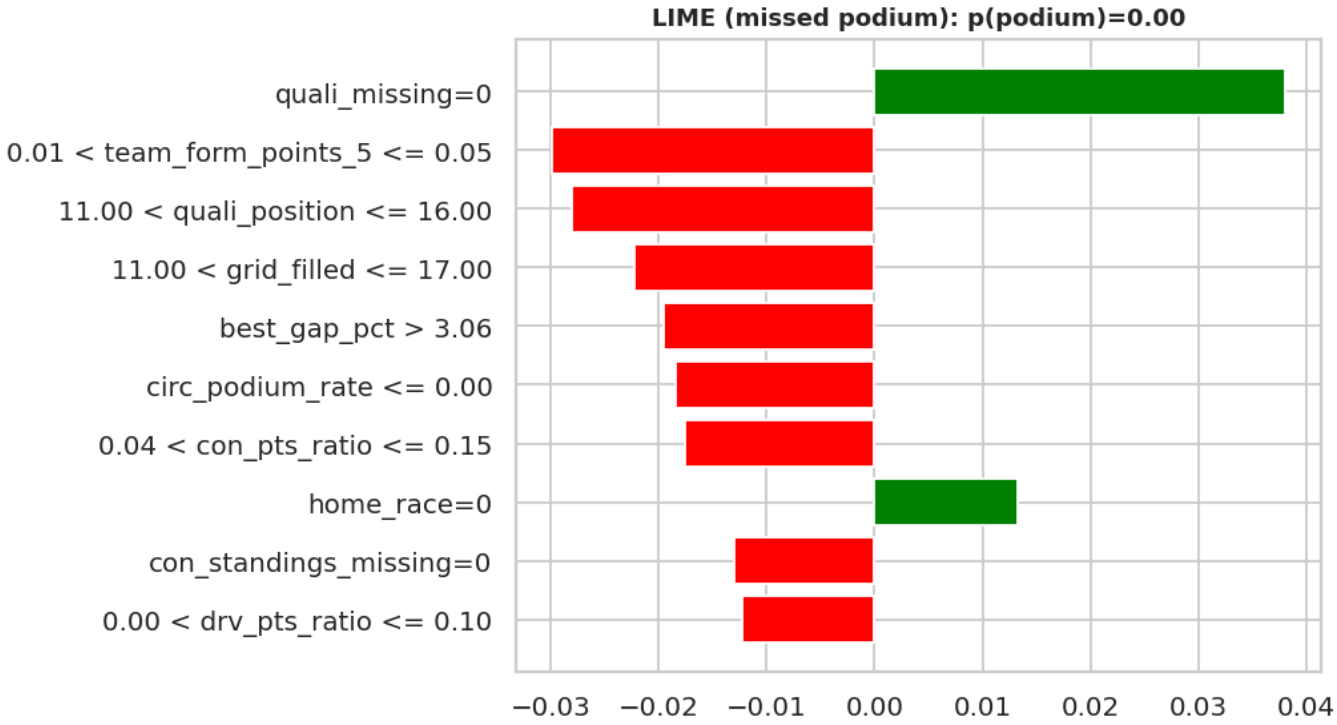


Figure 33 — lime missed *(29_lime_missed.png)*

8. Known limitations

- **Qualifying-coverage drift:** qualifying data are incomplete before ~2003; earlier seasons are excluded and EXP-E shows the effect of including them.
- **Era sensitivity / distribution shift:** regulation changes (2014 hybrid, 2022 ground effect) alter what predicts podiums; the test seasons are the newest regime.
- Podium prevalence is ≈3/field size and varies by era; thresholds tuned on validation may not transfer.
- Small per-circuit and per-constructor samples → wide intervals (Q1, Q3); rebrand lineage mapping is manual.
- Home flag relies on a manual nationality→country map; constructor strength is not controlled in Q2.
- Mechanical-retirement status classification is keyword-based and ambiguous for statuses like "Retired"/tyre damage (sensitivity reported).
- Race-day information (weather, safety cars, strategy, sprint results) is not available/used.
- **Feature pruning** is decided on validation only: Groups dropped (validation PR-AUC does not suffer without them, mean Δ ≥ 0.005): ['experience_context', 'driver_form']. Redundancy pruning not applied (mean Δ -0.0007, tolerance 0.005). Final feature set: 24 of 37 candidates.
- `is_pit_start`: in Ergast `grid=0` also covers non-starters, so the flag can carry a little outcome information; see the diagnostic in §8.
- **Partial records:** Inference behaviour: the complete record scores p=0.95; the same record without qualifying gaps, form and standings scores p=0.20; a record with only 'grid_filled=1' scores p=0.13. Probabilities fall as inputs disappear because remaining gaps are filled with training-set medians (a mid-field driver/team), so a partial record is a weaker, more conservative prediction — not a like-for-like one. Deterministic fields (front-row flag, pit-start flag, quali=grid / no penalty) are derived rather than imputed. Treat 'partial' outputs as low-confidence and prefer a missing-data-aware workflow (supply form/standings whenever available).
- Explanations describe model behavior, not causation.